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Effects of Nitrogen on the Microstructure and Mechanical Properties of Fe–28Mn–10Al–0.8C Low-Density Steel

Shuai Guo, Hangyu Zhu, Jixuan Zhao, Mingming Song, Jianli Li, and Zhengliang Xue*

Herein, the relationships among nitrogen content, inclusion characteristics, microstructure, and mechanical properties of low-density steel are investigated. The austenite grain size is refined with nitrogen addition to the low-density steel, and the ultimate tensile strength, plasticity, fracture toughness, and impact toughness are distinctly enhanced. However, the yield strength decreases due to a decrease in the ferrite content with nitrogen addition. AlN is determined to be the main inclusion in the high-nitrogen low-density steel, and the number percentage of AlN that is less than 2 μm reaches 84%, which promotes the refinement of austenite grains. In addition, aggregated AlN inclusions in the dimples of the steel are observed, which are generated during the smelting process, and the aggregated AlN inclusions tend to cause crack initiation in the low-density steel.

1. Introduction

In recent years, the rapid development of the automobile industry has resulted in problems related to energy consumption and air pollution. Accordingly, researchers have focused on developing lightweight materials for use in automobiles.[1,2] Ham et al.[3] developed Fe–34Mn–10Al–0.76C austenitic low-density steel. The density and weight of this Fe–Mn–Al–C low-density steel were later reduced by adding Al, Mn, and C (1.3% weight reduction per 1 wt% Al addition).[4] This low-density steel has outstanding mechanical properties, with an ultimate tensile strength (UTS) of 0.6–2.0 GPa and elongation of 20–100%.[5–10]

Many researchers have confirmed that nitrogen can improve the mechanical properties of various steels.[11–18] Gu and Wang et al.[11,15] reported that nitrogen could promote the precipitation of V(C, N) and (Ti, Nb)(C, N) and could refine the grains of Cr–Mo–V die steel and pipeline steel. Feng et al.[16] concluded that the addition of nitrogen inhibited the formation of δ-ferrite and refined the prior austenite grains in high-nitrogen martensitic stainless steel. The effects of nitrogen on the microstructures and corrosion mechanisms have also been discussed. Zhang et al.[17] investigated the relationships among nitrogen content, microstructure, and mechanical properties of S32654 superaustenitic stainless steel and confirmed that nitrogen could improve the tensile strength and elongation of S32654 steel. Jiao et al.[18] found that the mechanical properties of as-cast M42 high-speed steel were enhanced by the addition of 0.205 wt% nitrogen due to an increase in the retained austenite content and the transformation of MC to M(C, N). Nevertheless, few studies have investigated the effects of nitrogen addition on the mechanical properties of low-density steel.

The addition of nitrogen to molten low-density steel may lead to the formation of nitride inclusions. Lu et al.[19] reported that AlN inclusions form in liquidus steel when the contents of Al and N exceed the critical solubility of AlN inclusions in high-nitrogen stainless bearing steel. Alba et al.[20] reported that AlN was the dominant inclusion, and the total number of inclusions increased with an increase in the nitrogen content in Fe–5Mn–3Al steel. Park et al.[21] reported that both AlN and AlON inclusions increased with increasing Al content in Fe–Mn–Al steel. In our previous study,[22] we found that AlN inclusions were generated in molten steel and then agglomerated in Fe–23Mn–*x*Al–0.7C low-density steel. Therefore, the effects of AlN inclusion on the microstructure and mechanical properties of Fe–Mn–Al–C steel must be further studied.

In this study, the effects of nitrogen on the microstructure and mechanical properties of Fe–28Mn–10Al–0.8C (wt%) low-density steel are investigated. The relationships among nitrogen content, inclusion characteristics, microstructure, and mechanical properties are also discussed.

2. Experimental Section

2.1. Materials Preparation

Low-density steels with a chemical composition of Fe-28 wt% Mn-10 wt% Al-0.8 wt% C and different nitrogen contents were used to investigate the effects of nitrogen on their microstructure and mechanical properties. The raw materials for the steels included pure iron, ferromanganese, carbon particles, aluminum

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Table 1. Chemical compositions of the investigated low-density steels (wt%).

Steel	Mn	Al	C	N	S	Fe
LN	28.21	9.90	0.82	0.0025	0.0085	Bal.
HN	28.10	9.95	0.81	0.1326	0.0093	Bal.

particles, and manganese nitrogen alloy. Ingots (50 kg) of low-density steel were manufactured using a vacuum medium-frequency induction furnace. The chemical compositions of the low-density steels are listed in **Table 1**. The ingots were held at 1423 K for 2 h and then multidirectionally hot forged in a temperature range of 1123–1323 K, followed by air cooling.

2.2. Procedure

The machining dimensions of the tensile and impact steel samples are shown in **Figure 1**. Tensile tests were conducted at room temperature using a universal material testing machine (WDW-200) at a strain rate of $1 \times 10^{-3}\ s^{-1}$ and tensile speed of $3\ mm\ min^{-1}$. The yield strength (YS), UTS, reduction of area (RA), and elongation (A) were determined from the tensile test, and then the stress–strain curves of the tensile test were obtained. The stress–strain curves were corrected by adding the real elastic deformation to plasticity deformation obtained from the tensile test. The impact toughness was tested using an impact testing machine (JB-300B) to obtain the impact energy (CVN) of the steel samples.

2.3. Analysis

The samples for microstructural observation were prepared through mechanical grinding, followed by corroding with a solution of 96 vol% alcohol and 4 vol% nitric acid. Fracture samples following tensile and impact tests were observed using a scanning electron microscope (SEM) and a superdepth of field microscope (VHX-500). The morphologies and compositions of the inclusions were observed using an SEM equipped with an energy-dispersive spectroscope (EDS). 50 random view fields (500× magnification) were obtained using SEM. The number and size of the inclusions were measured using ImageJ software.

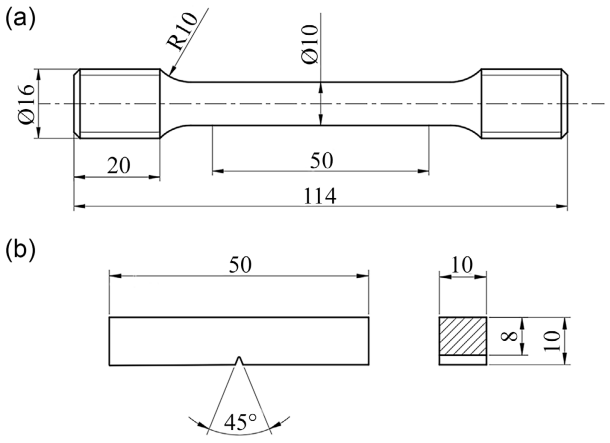


Figure 1. Schematic of specimens for a) tensile and b) impact tests.

3. Results

3.1. Mechanical Properties

The mechanical properties of the investigated low-density steels are presented in **Figure 2** and **Table 2**, which show that the values of UTS for the low-nitrogen (LN) and high-nitrogen (HN) steels were 832.5 and 841.5 MPa, respectively. The RA and A values for the HN steel were 13% and 12% higher, respectively, than those of the LN steel. In addition, the impact energy of the HN steel was 130.5 J, which was 49.5 J higher than that of the LN steel. Notably, the YS of the HN steel was significantly lower than that of the LN steel. With the addition of nitrogen, the UTS, plasticity, and impact toughness were enhanced, and the YS was reduced.

3.2. Microstructure

Figure 3 shows the microstructures of the LN and HN steels. It can be seen that the mean grain sizes of the LN and HN steels were 20.5 and 12.1 μm, respectively. Finer grains were obtained for HN steel. In addition, ferrites with long parallel stripes were observed in the boundaries of austenite matrix due to the low-precipitation temperature of ferrite. The ferrite in the LN steel was coarser and larger than that in the HN steel. The area percentage of ferrite in the LN steel was 5.08% as compared with 1.84% in the HN steel. The results indicated that nitrogen addition could promote grain refinement and an increase in the area

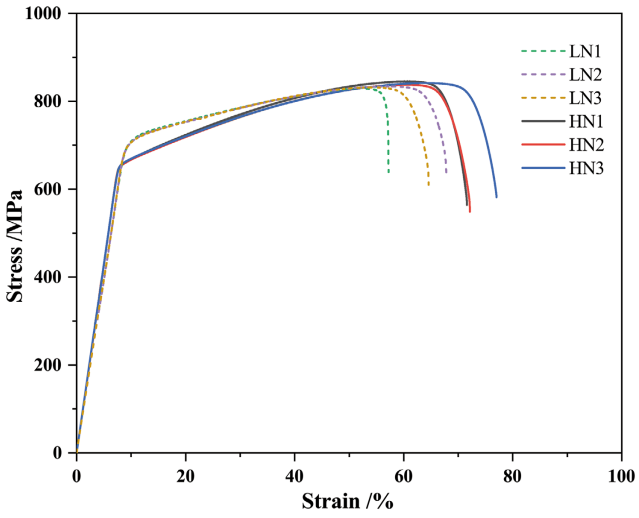


Figure 2. Stress–strain curves for the LN and HN steel samples.

Table 2. Mechanical properties of the experimental steel samples.

Steel	YS [MPa]	UTS [MPa]	RA [%]	A [%]	CVN [J]
LN	685(±1.5)	832.5(±1.5)	57(±6)	62.5(±5)	81(±3)
HN	647(±1)	841.5(±3.5)	70(±2)	74.5(±2.5)	130.5(±5.5)

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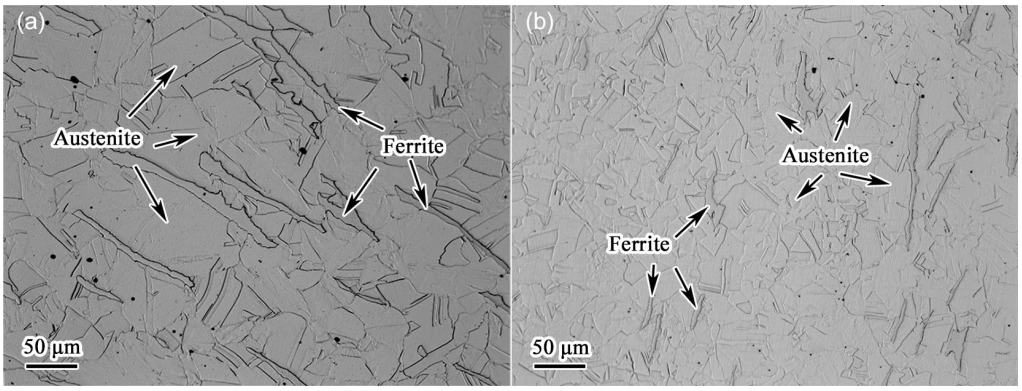


Figure 3. Microstructures of a) LN and b) HN steels.

ratio of austenite in Fe–Mn–Al–C low-density steel. In addition, more fine deformation twins were detected in the austenite grains of the HN steel after forging.

3.3. Characteristics of Inclusions

Figure 4, **5** show the morphologies of typical inclusions in LN and HN steels. The dominant inclusions in the LN steel were AlN, Mn(Mg)S, and their composite inclusions. The sulfide inclusion was elliptical with a smooth edge, and the AlN inclusion had a polyhedral structure. Sulfide inclusions occasionally precipitated on the surface of AlN to form AlN–Mn(Mg)S complex inclusions. For the HN steel, the main inclusion was AlN with a straight edge and sharp angle. It is worth noting that AlN inclusions could be broken after forging (see Figure 5b), which generally became the crack origin of the steel fracture.[23]

Figure 6, **7** show the statistical results of the numbers and sizes of inclusions in the LN and HN steels. The number percentages (NPs) of inclusions less than 4 μm in size in the LN and HN steels were 78% and 95%, respectively. The NP value of inclusions (mainly AIN) less than 2 μm in size in the HN steel reached 84%. The total number of inclusions in the HN steel was significantly higher than that in the LN steel, particularly for inclusions less than 2 μm in size. The numbers, sizes, and area ratios of the inclusions are listed in **Table 3**. The number densities were 133 and 73 mm^{-2} for the HN and LN steels, respectively. Nitrogen addition could increase the number and area ratio of inclusions while decreasing the average size in low-density steels.

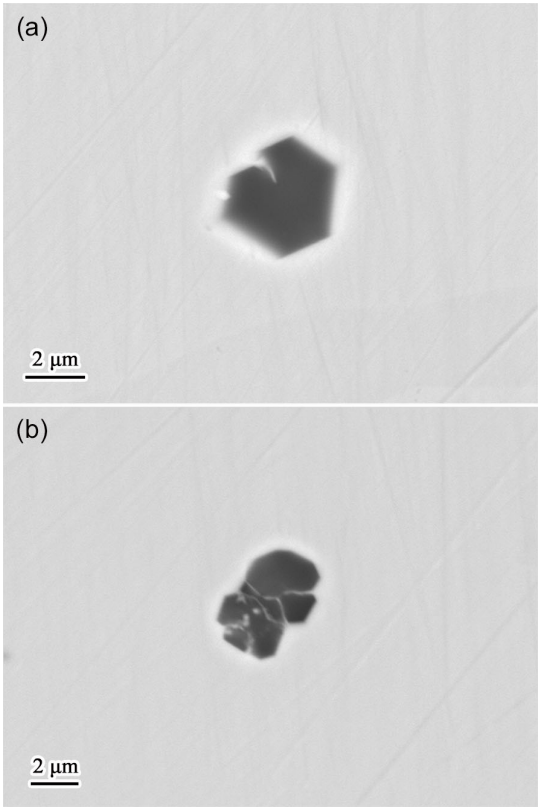


Figure 5. Typical morphologies of AlN inclusions in HN steel.

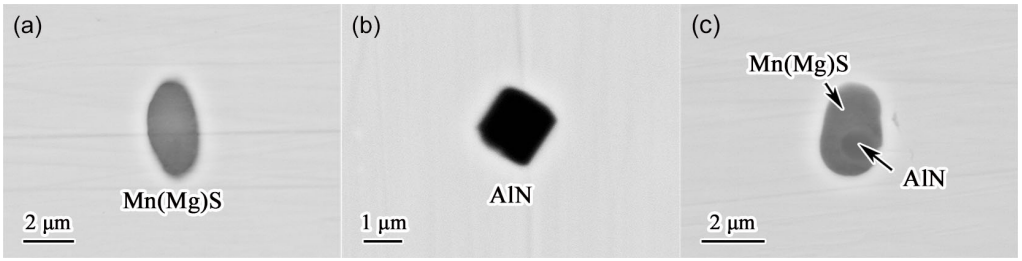


Figure 4. Typical morphologies of inclusions in LN steel.

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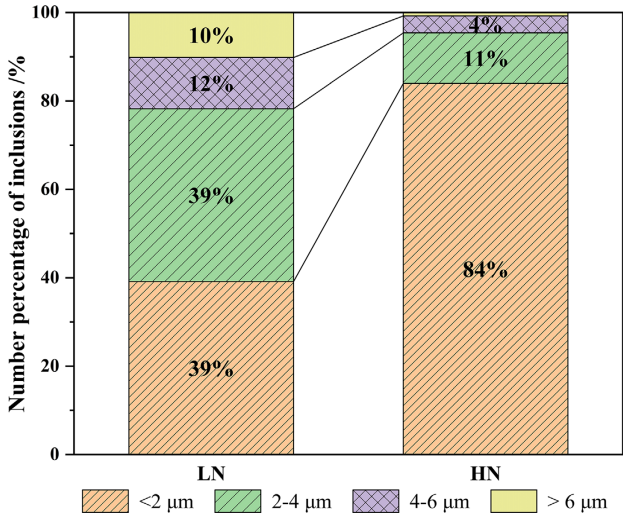


Figure 6. NPs of inclusions in LN and HN steels.

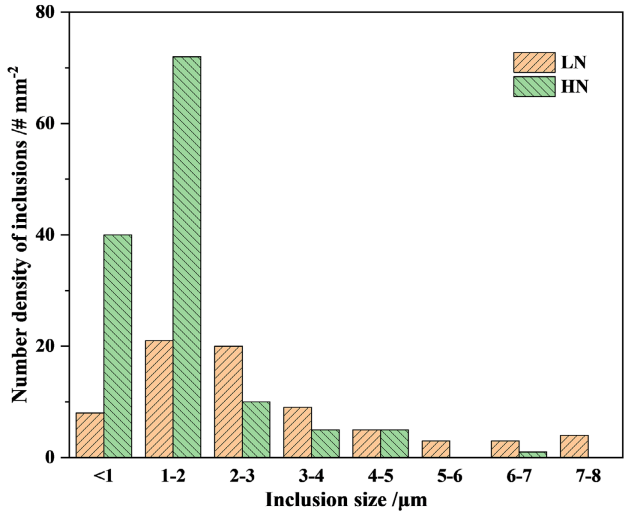


Figure 7. Number densities of inclusions in LN and HN steels.

Table 3. Numbers, sizes, and area ratios of nonmetallic inclusions.

Steel	Number density [# mm]$^{-2}$	Average size [μm]	Area ratio [%]
LN	73	3.435	0.212
HN	133	1.521	1.005

4. Discussion

4.1. Effects of Nitrogen on Inclusions

The addition of nitrogen promoted the formation of AlN in the low-density steels investigated in this study. The formation of AlN inclusions was mainly controlled by the reaction equilibrium based on Al and N content, temperature, Gibbs free energy, activity coefficients, and interaction coefficients. Based on the calculation method reported in our previous work,[22] an AlN stability diagram of

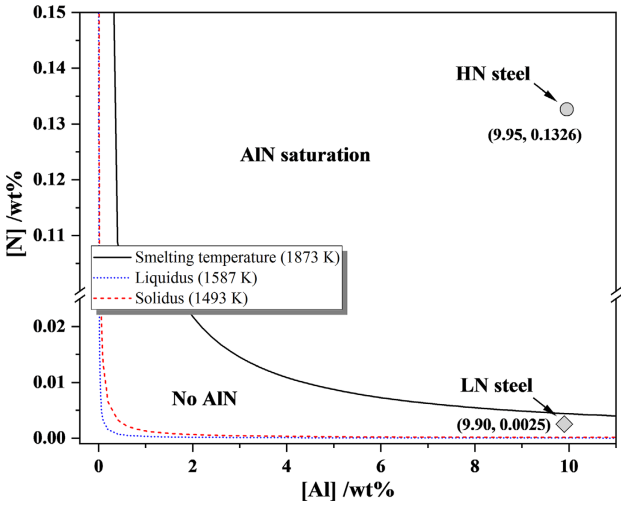


Figure 8. AlN stability diagram of Fe–28Mn–10Al–0.8C steels.

Fe–28Mn–10Al–0.8C steels was calculated and is presented as **Figure 8**. In addition, the liquidus (1587 K) and solidus (1493 K) of the investigated steel were calculated using FactSage 8.0. The Al and N contents of the LN and HN steels are also indicated in Figure 8, which can be used to examine the formation of AlN inclusions (i.e., by comparison with the stability diagram). For LN steel, the composition of Al and N fell below the equilibrium line of the smelting temperature, which suggests that AlN was not a stable phase and could not be generated during the smelting process. However, for HN steel with 0.1326 wt% nitrogen, the composition far exceeded the equilibrium line. The results indicate that AlN inclusions can be easily formed and stably exist during the smelting process.

In general, AlN inclusions are generated during the solidification process with lower Al and N contents and can be formed in molten steel with higher Al and N contents. Alba et al.[20] concluded that the main inclusion was AlN–MnS inclusion in Fe–5Mn–3Al steel, whereas the nitrogen content was less than 10 ppm. With an increase in N content, AlN became the dominant inclusion. The results coincided with those observed in the current work, as shown in Figure 4–7.

4.2. Effects of Nitrogen on Microstructure

To clarify the effects of nitrogen content on the microstructure of low-density steel, a pseudobinary phase diagram of the Fe–28Mn–10Al–0.8C–N system was calculated using Thermo-Calc software based on the TCFE8 database (see **Figure 9**). The figure shows that the maximum formation temperature and temperature range of the austenite phase increased significantly and that ferrite gradually disappeared with the increase in nitrogen content in the steel. As a stabilizing element for austenite, nitrogen expanded the austenite phase area and promoted the growth of the austenite nucleus, thereby preventing the growth of ferrite. As shown in Figure 3, a large number of annealing twins were observed between the austenite grains, which crossed each other and cut the matrix to refine the grains.[24]

The differences in the numbers and sizes of austenite grains between the HN and LN steels were associated with the AlN inclusions. It is well known that dispersive and fine second-phase

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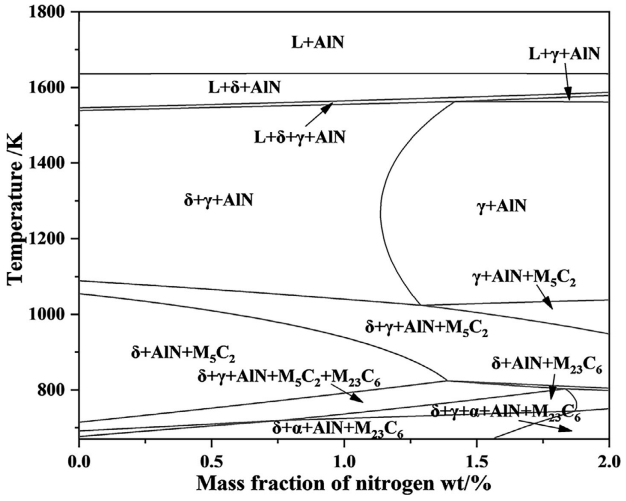


Figure 9. Pseudobinary diagram of the Fe–28Mn–10Al–0.8C–N system.

particles refine the grains in steel.[11,14–25] As previously mentioned, the number of inclusions smaller than 2 μm in the HN steel was significantly greater than that in the LN steel, and AlN inclusions were formed during the smelting process for the HN steel. The fine AlN inclusions acted as the cores of the heterogeneous nucleation of austenite during the solidification process.[26] In addition, Li et al.[27,28] reported that some fine and dispersed inclusions could effectively pin the austenite grain boundary and prevent the growth of austenite grains.

4.3. Effects of Nitrogen on Mechanical Properties

Researchers have confirmed that steel with fine grains has excellent strength, plasticity, and toughness.[11–18,28] The results of the tensile tests in the present study were consistent with those from the literatures, as indicated in Table 2 and Figure 2. The results showed that the plasticity of the HN steel was obviously better than that of the LN steel, and the LN steel generally broke earlier than the HN steel during local necking. Hans and Roylance et al.[29,30] reported that the areas under stress–strain curves can represent the rated energy consumed by the fracture metal materials. According to the theory, the energy consumption of HN steel fracture is higher than that of LN steel. **Figure 10** shows the morphologies of the tensile fractures of the LN and HN steels. The macroscopic necking degree of the HN steel was considerably greater than that of the LN steel. Large dimples were observed in the fractures of the HN steel, which was usually surrounded by small dimples. The dimples in the LN steel were smaller in size and greater in number, which indicated that the local stress concentration of the LN steel was more severe during the deformation process. The HN steel was larger and had a deeper depth and more uniform distribution of dimples, which suggested that the plasticity and toughness were improved.[31] The improved plasticity and fracture toughness in the HN steel was due to the fine grains, and many twins caused the deformation to become more uniform and relieve the stress concentration of the deformed steel matrix. In addition, many grain and twin boundaries prevented crack propagation and enhanced fracture toughness.[32,33] Griffith's equation describes the effect of grain refinement on toughness, as shown in the following equation.[28]

$$\sigma_f = \left[\frac{4E\gamma}{\pi(1-\nu^2)c_0}\right]^{1/2} \quad (1)$$

where σ_f is the fracture stress, E is Young's modulus, γ is the plastic deformation energy during crack propagation, ν is Poisson's ratio, and c_0 is the effective grain size.

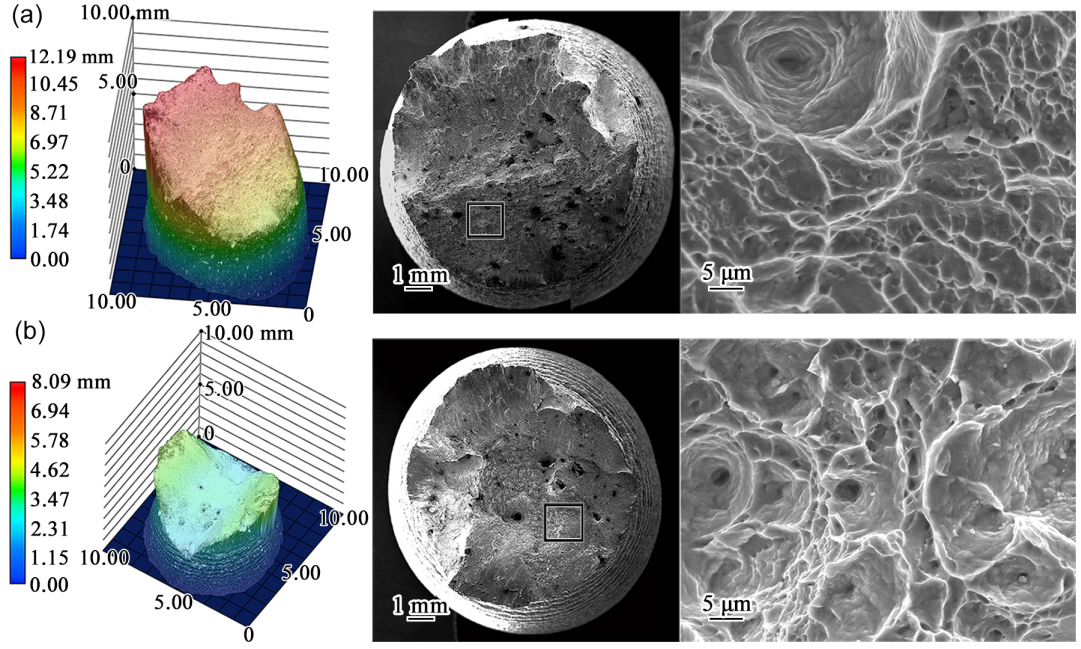


Figure 10. Morphologies of tensile fracture for a) LN and b) HN steels.

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In addition, the fracture morphology of the HN steel had a crater shape, whereas the fracture section of the LN steel was 45° to the load direction, as shown in Figure 10. The results revealed that the LN steel sample was fractured by shear stress at an angle of 45° before the stress in the loading direction reached its tensile strength. Therefore, the tensile strength of the LN steel was lower than that of HN steel. The results were derived from the many grain and twin boundaries in the steel matrix. The grain and twin boundaries also hindered the dislocation movement in the HN steel, thereby indirectly promoting the dislocation strengthening effect and further enhancing the strength and hardness of the HN steel.

The Hall–Page formula displays the relationship between the YS and grain size, as shown in Equation (2).[34]

$$\sigma_y = \sigma_0 + \frac{K_y}{d^{-1/2}} \tag{2}$$

where σ_s is the YS of polycrystalline materials, σ_0 is the lattice friction resistance when a single dislocation is moved, d is the average grain diameter, and K_y represents the influence degree of grain boundaries on strength. The YS of polycrystalline materials increased with a decrease in the grain size. In the present study, the increase in nitrogen content led to grain refinement in the HN steel, which explains the improved fracture toughness, tensile strength, and plasticity of the HN steel. However, the YS of the HN steel was slightly lower than that of the LN steel with values of 647 and 685 MPa, respectively, as shown in Table 2. This may have resulted from the different contents of ferrite in the HN and LN steels. The area percentages of ferrite in the LN and HN steels were 5.08% and 1.84%, respectively (see Figure 3). Chen et al.[5] concluded that the YS of steels can be improved by increasing the ferrite content in austenite-based duplex steels.

Figure 11 shows the morphologies of the impact fractures of LN and HN steels. In general, fine grains and twins considerably improve the impact toughness of steel.[15,35] In our study, the grain boundaries of metal materials with fine grains were more curved in the HN steel, which could prevent crack propagation and absorb more impact energy during the impact test. Meanwhile, the ferrite precipitated at the grain boundary of austenite matrix contributes to crack initiation and brittle crack propagation.[36,37] The impact fractures of the LN and HN steel samples presented a standard ductile impact fracture, which included the fibrous, radiation, and shear lip zones. The area ratio of the shear lip zone in the HN steel sample was significantly higher than that in the LN steel, indicating that the crack growth rate of the HN steel was slower and the impact toughness was better. Figure 11a shows that the fracture dimples of the LN steel were elongated in the direction of crack propagation, which was greatly affected by the shear force, resulting in accelerated crack propagation. Figure 11b shows that the size and average depth of the fracture dimples of the HN steel were large and uniform. The HN steel matrix exhibited higher deformation and absorbed more energy during the impact test.

Finally, the changes in the number and sizes of inclusions derived from nitrogen addition also had obvious effects on the mechanical properties of the HN steel. Different types of inclusions were detected in the dimples on the fracture surfaces of the test samples, as shown in **Figure 12**, indicating that these inclusions became the initial point of fracture. The sulfide inclusions showed good deformation ability and were typically large; however, these sulfide inclusions could easily break away from cracks at the inclusion–matrix interface due to their weak binding force to the steel matrix.[38,39] In comparison, the brittle AlN inclusions had a high melting point and exhibited poor deformation ability. For HN steel, fine and dispersed AlN inclusions could refine the austenite grains and improve their mechanical properties.

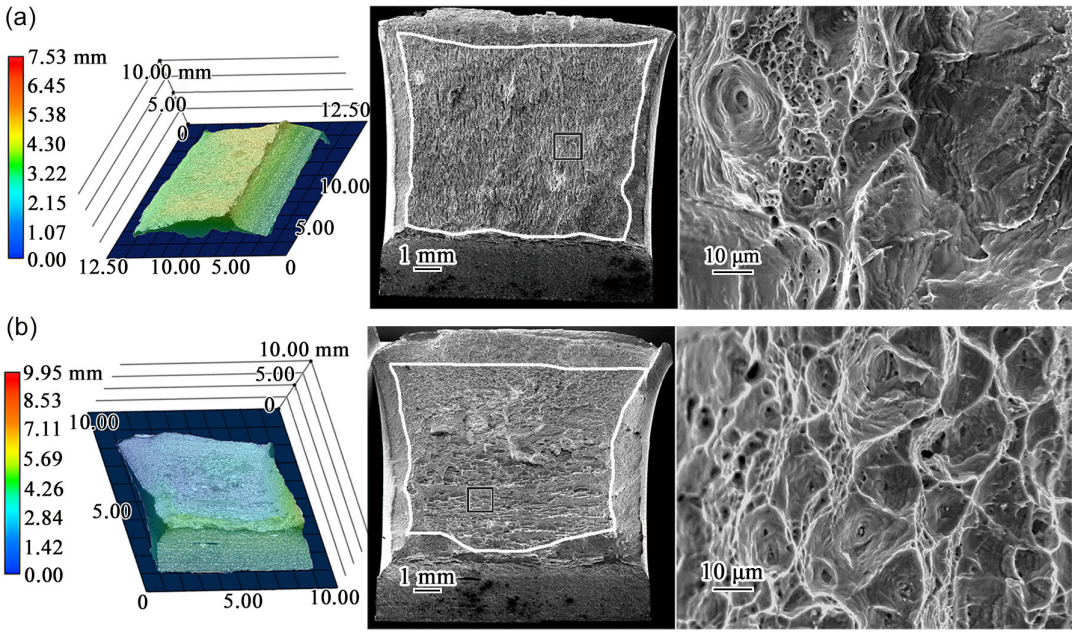


Figure 11. Morphologies of impact fracture for a) LN and b) HN steels.

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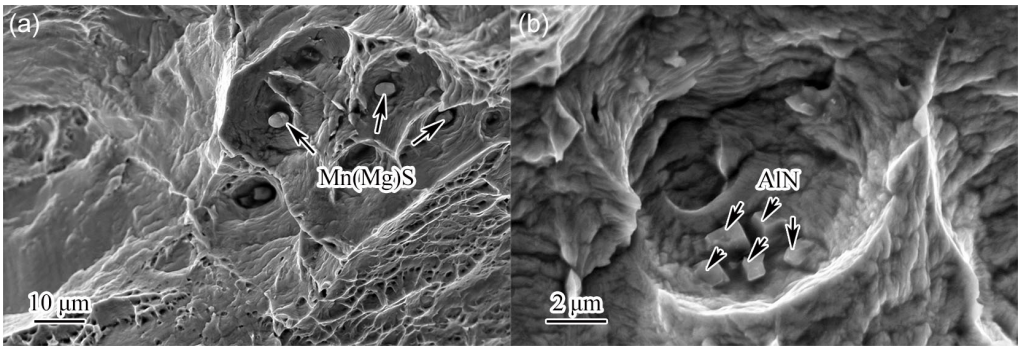


Figure 12. Typical inclusions in the dimples of a fracture surface.

However, aggregated AlN inclusions were detected in the HN steel (see Figure 12b), which could deteriorate the steel properties. Wang et al.[22] also observed aggregated AlN inclusions in low-density steel and concluded that aggregated AlN inclusions were generated during the smelting process due to the combined effects of the cavity bridge and viscous resistance.

5. Conclusion

In this study, the effects of nitrogen on the non-metallic inclusions, microstructure, and mechanical properties of Fe-28Mn-10Al-0.8C low-density steel were investigated. The following conclusions were drawn. 1) Nitrogen addition significantly decreased the grain size and increased the grain number and austenite content, which improved the UTS, plasticity, fracture toughness, and impact toughness of low-density steels. However, the YS decreased with a decrease in the ferrite content with nitrogen addition. 2) The main inclusions in the LN steel were AlN, sulfide, and their compound inclusions. The dominant inclusion in the HN steel was AlN, and the NP of AlN less than 2 μm in size in the HN steel reached 84%. The fine and dispersed AlN inclusions that were small in size could refine the austenite grains, which is beneficial to plasticity and toughness. 3) Aggregated AlN inclusions were detected in the dimples of the HN steel that were generated during the smelting process. The aggregated AlN inclusions were detrimental to the mechanical properties of the low-density steel, which tended to initiate crack formation.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

AlN inclusions, Fe–Mn–Al–C, high-nitrogen steels, low-density steels, mechanical properties, microstructures, nonmetallic inclusions

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[1] J. H. Schmitt, T. Iung, *C. R. Phys.* **2018**, *19*, 641.
[2] K. K. Dama, V. S. Babu, R. N. Rao, *Mater. Today: Proc.* **2018**, *5*, 20966.
[3] J. L. Ham, R. E. Cairns Jr, *Prod. Eng.* **1958**, *29*, 50.
[4] G. Frommeyer, U. Brüx, *Steel Res. Int.* **2006**, *77*, 627.
[5] S. Chen, R. Rana, A. Haldar, R. K. Ray, *Prog. Mater. Sci.* **2017**, *89*, 345.
[6] J. D. Yoo, K. T. Park, *Mater. Sci. Eng. A* **2008**, *496*, 417.
[7] H. Kim, D. W. Suh, N. J. Kim, *Sci. Technol. Adv. Mater.* **2013**, *14*, 14205.
[8] J. Pang, Z. Zhou, Z. Zhao, D. Tang, J. Liang, Q. He, *Metals* **2019**, *9*, 897.
[9] Z. C. Li, H. Ding, R. D. K. Misra, Z. H. Cai, H. X. Li, *Mater. Sci. Eng. A* **2016**, *672*, 161.
[10] J. Moon, S. J. Park, J. H. Jang, T. H. Lee, C. H. Lee, H. U. Hong, H. N. Han, J. Lee, B. H. Lee, C. Lee, *Acta Mater.* **2018**, *147*, 226.
[11] J. Gu, J. Li, R. Chang, L. Li, *Mater. Sci. Eng. A* **2019**, *766*, 138386.
[12] T. V. Rashev, A. V. Eliseev, L. T. Zhekova, P. V. Bogev, *Steel Trans.* **2019**, *49*, 433.
[13] J. Burja, M. Koležnik, Š. Župerl, G. Klančnik, *Mater. Technol.* **2019**, *53*, 919.
[14] P. Behjati, A. Kermanpur, A. Najafizadeh, H. S. Baghbadorani, L. P. Karjalainen, J. Jun, Y. Lee, *Metall. Mater. Trans. A* **2014**, *45*, 6317.
[15] F. H. Wang, Y. Q. Chen, Q. L. Wang, S. K. Mou, *Trans. Mater. Heat Treat.* **2020**, *41*, 106.
[16] H. Feng, Z. Jiang, H. Li, P. Lu, S. Zhang, H. Zhu, B. Zhang, T. Zhang, D. Xu, Z. Chen, *Corros. Sci.* **2018**, *144*, 288.
[17] S. Zhang, H. Li, Z. Jiang, Z. Li, J. Wu, B. Zhang, F. Duan, H. Feng, H. Zhu, *J. Mater. Sci. Technol.* **2020**, *42*, 143.
[18] W. Jiao, H. Li, H. Feng, Z. Jiang, J. Dai, H. Zhu, S. Zhang, M. Chu, W. Wu, *ISIJ Int.* **2020**, *60*, 564.
[19] P. Lu, H. Li, H. Feng, Z. Jiang, H. Zhu, Z. Liu, T. He, *Metall. Mater. Trans. B* **2021**, *52*, 2210.
[20] M. Alba, M. Nabeel, N. Dogan, *Crystals* **2020**, *10*, 836.
[21] J. H. Park, D. J. Kim, D. J. Min, *Metall. Mater. Trans. A* **2012**, *43*, 2316.
[22] W. Wang, H. Zhu, Y. Han, J. Li, Z. Xue, *Ironmaking Steelmaking* **2021**, *48*, 1038.
[23] H. Abdesselam, J. Crépin, A. Pineau, A. L. Rouffie, P. Gaborit, L. Menut-Tournadre, T. F. Morgeneyer, *Int. J. Fatigue* **2018**, *106*, 132.

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[24] G. Song, Z. Wang, Y. Zheng, T. Zhang, *AIP Conf. Proc.* **2019**, *2073*, 20027.
[25] H. Guo, S. Yang, T. Wang, H. Yuan, Y. Zhang, J. Li, *J. Mater. Sci. Technol.* **2021**, *99*, 277.
[26] X. Zhao, Z. Yang, F. Zhang, X. Long, C. Chen, *Mater. Sci. Technol.* **2019**, *35*, 147.
[27] F. Li, H. Li, Y. Wu, D. Zhao, B. Peng, H. Huang, S. Zheng, J. You, *J. Mater. Res.* **2017**, *32*, 2307.
[28] F. Zhao, M. Wu, B. Jiang, C. Zhang, J. Xie, Y. Liu, *Mater. Sci. Eng. A* **2018**, *731*, 360.
[29] B. Hans, V. G. Gavriljuk. *Key Eng. Mater.* **2007**, *345*, 421.
[30] D. Roylance, Stress-strain curves, Massachusetts Institute of Technology Study, Cambridge **2001**.
[31] Q. Long, J. Lu, T. Fang, *Mater. Sci. Eng. A* **2019**, *751*, 42.
[32] X. Wang, H. Wu, Y. Gu, L. Xu, P. Zhang, Q. Tang, *J. Iron Steel Res. Int.* **2021**, *27*, 1530.
[33] B. Wu, S. Huang, Z. Wang, J. Zhao, C. Shang, R. D. K. Misra, *Mater. Lett.* **2019**, *250*, 55.
[34] C. Wang, M. Wang, J. Shi, W. Hun, H. Dong, *J. Mater. Sci. Technol.* **2007**, *23*, 659.
[35] J. Wang, C. Enloe, J. Singh, C. Horvath, *SAE Int. J. Mater. Manuf.* **2016**, *9*, 488.
[36] Z. Huang, A. Hou, Y. Jiang, P. Wang, Q. Shi, Q. Hou. *J. Iron Steel Res. Int.* **2017**, *24*, 1190.
[37] S. Das, M. Mukherjee, T. K. Pal. *Eng. Failure Anal.* **2015**, *54*, 102.
[38] S. B. Hosseini, C. Temmel, B. Karlsson, N. G. Ingesten, *Metall. Mater. Trans. A* **2007**, *38*, 982.
[39] M. Wang, Y. Zeng, X. Wang, X. Xie, H. Fan, *J. Iron Steel Res.* **2008**, *20*, 46.

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